Conditional random fields

Highlights

- CRF is a form of undirected graphical model
- Proposed by Lafferty, McCallum and Pereira in 2001
- Used in many NLP tasks: e.g., Named-entity detection
- Types:
  - Linear-chain CRF
  - Skip-chain CRF
  - General CRF

Outline

- Graphical models
- Linear-chain CRF
- Skip-chain CRF

Graphical models

Graphical model

- A graphical model is a probabilistic model for which a graph denotes the conditional independence structure between random variables:
  - Nodes: random variables
  - Edges: dependency relation between random variables

- Types of graphical models:
  - Bayesian network: directed acyclic graph (DAG)
  - Markov random fields: undirected graph

Bayesian network

Bayesian network

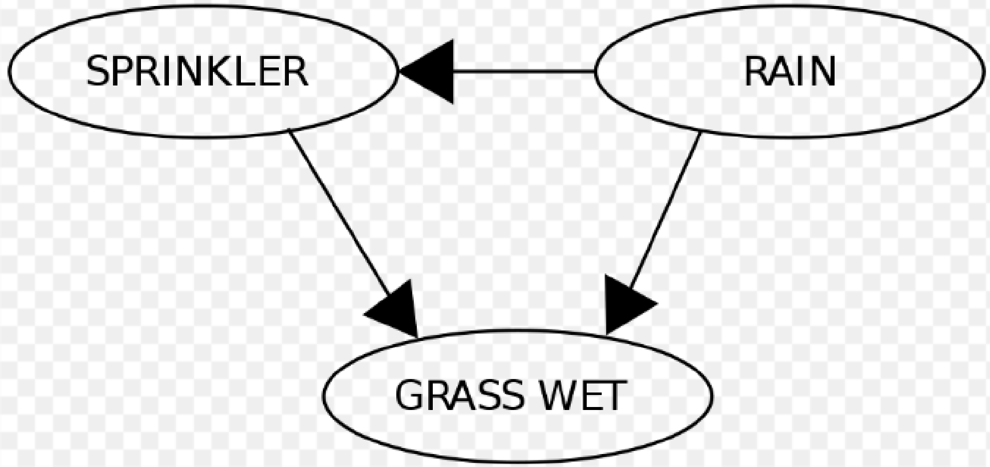
- Graph: directed acyclic graph (DAG)
  - Nodes: random variables
  - Edges: conditional dependencies
  - Each node X is associated with a probability function P(X | parents(X))

- Learning and inference: efficient algorithms exist.

An example

(from http://en.wikipedia.org/wiki/Bayesian_network)

	SPRINKLER	
RAIN	T	F
F	0.4	0.6
T	0.01	0.99



RAIN	
T	F
0.2	0.8

Pr(G, S, R)= Pr(G|S,R) Pr(S|R)Pr(R)

		GRASS WET	
SPRINKLER	RAIN	T	F
F	F	0.0	1.0
F	T	0.8	0.2
T	F	0.9	0.1
T	T	0.99	0.01

Pr(S|R)

Pr(G|S, R)

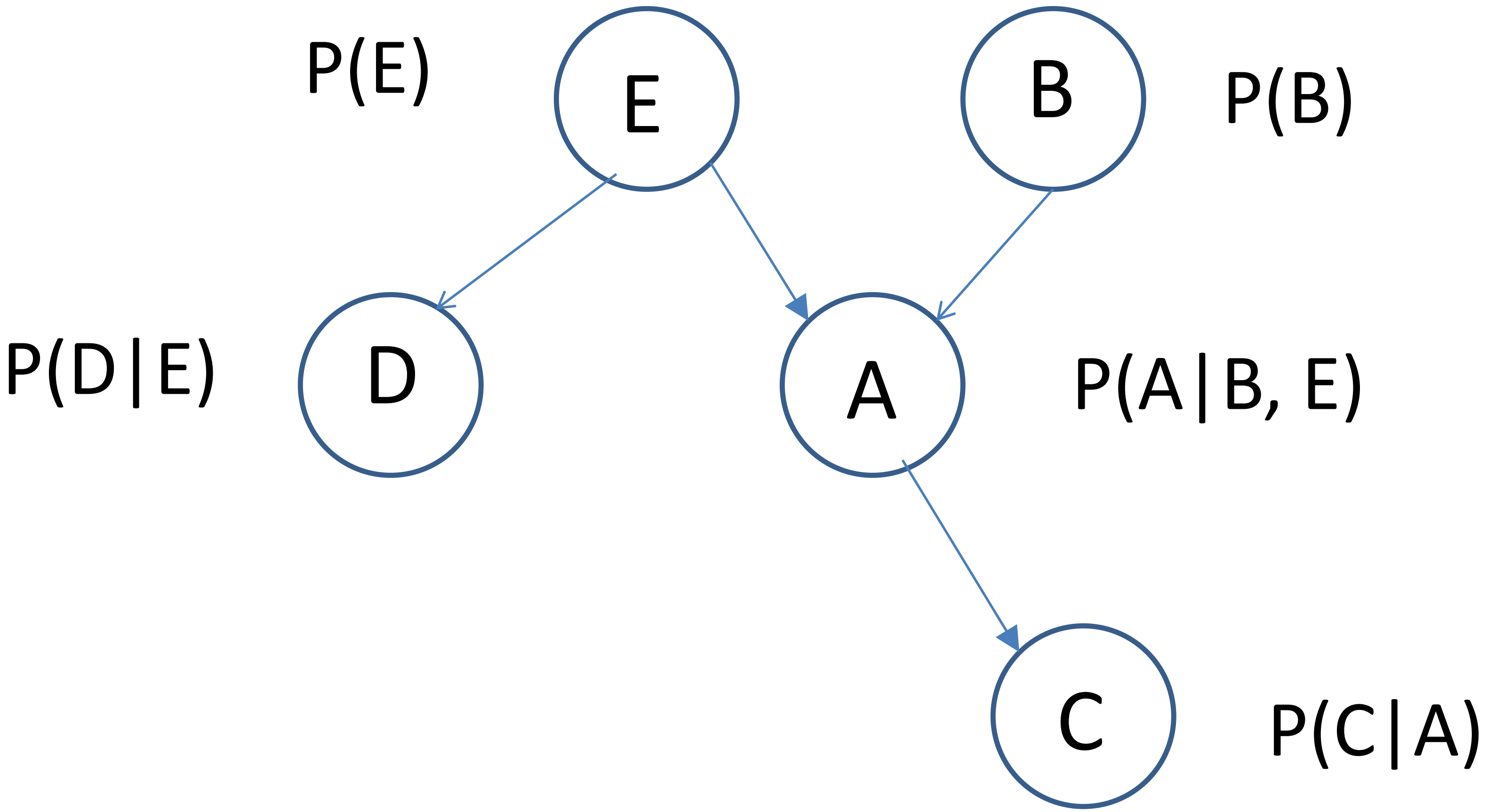
- What is the probability that it is raining, given the grass is wet?

$$\Pr(R=T \mid G=T) = \frac{\Pr(G=T, R=T)}{\Pr(G=T)} = \frac{\sum_{x\in\{T,F\}} \Pr(G=T, S=x, R=T)}{\sum_{x,y\in\{T,F\}} \Pr(G=T, S=x, R=y)}$$

$$\begin{aligned}\Pr(G=T, S=T, R=T) &= \Pr(G=T \mid S=T, R=T)\Pr(S=T \mid R=T)\Pr(R=T) \\ &= 0.99 \times 0.01 \times 0.2 \\ &= 0.00198.\end{aligned}$$

$$\Pr(R=T \mid G=T) = \frac{0.00198_{TTT} + 0.1584_{TFT}}{0.00198_{TTT} + 0.288_{TTF} + 0.1584_{TFT} + 0.0_{TFF}} = \frac{891}{2491} \approx 35.77\%.$$

Another example

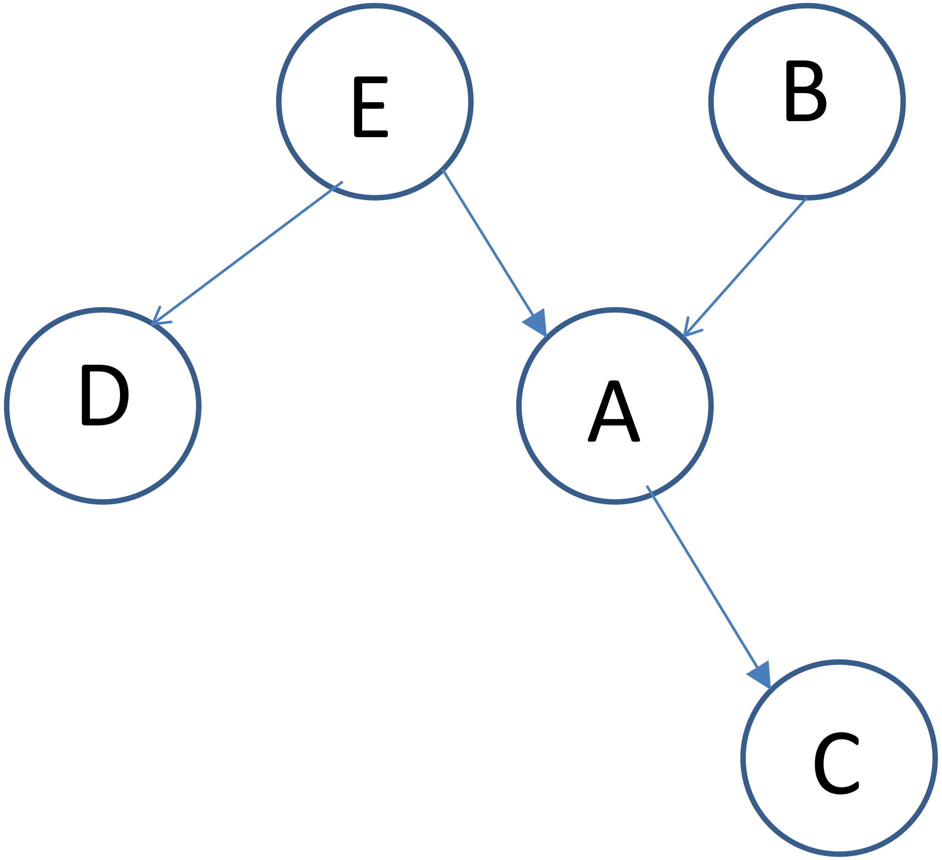


Bayesian network: properties

Local Markov property: each variable X_i is conditionally independent of its nondecendants given its parents variables.

$$P(X_1, ..., X_n)$$
$$= \prod_{i=1}^{n} P(X_i | X_1, ..., X_{i-1}))$$

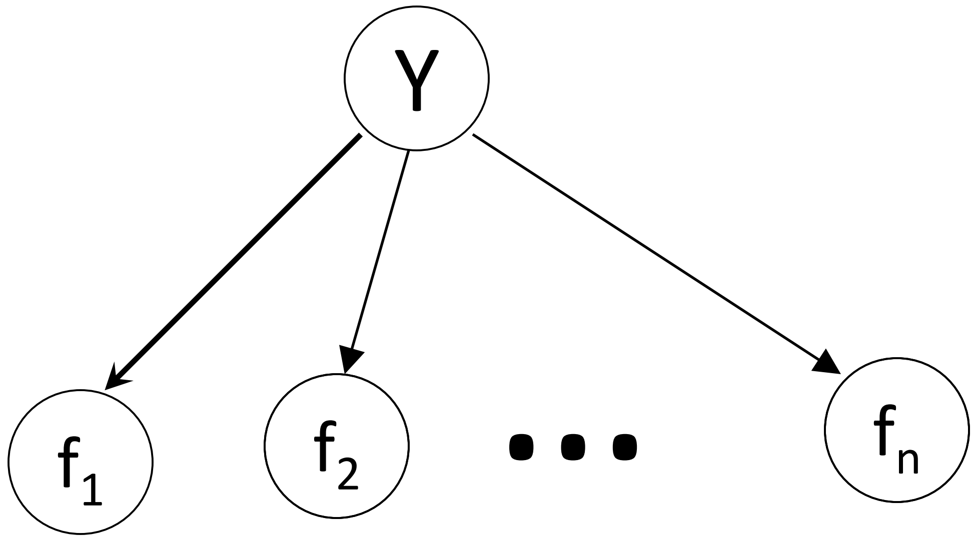
$$= \prod_{i=1}^{n} P(X_i | parents(X_i))$$



$$P(A,B,C,D,E) = P(B)P(E)P(A|B,E)P(C|A)P(D|E)$$

$$P(B,E|C,D) = \frac{P(B,E,C,D)}{P(C,D)} = \frac{\sum_A P(A,B,C,D,E)}{\sum_A \sum_B \sum_E P(A,B,C,D,E)}$$

Naïve Bayes Model

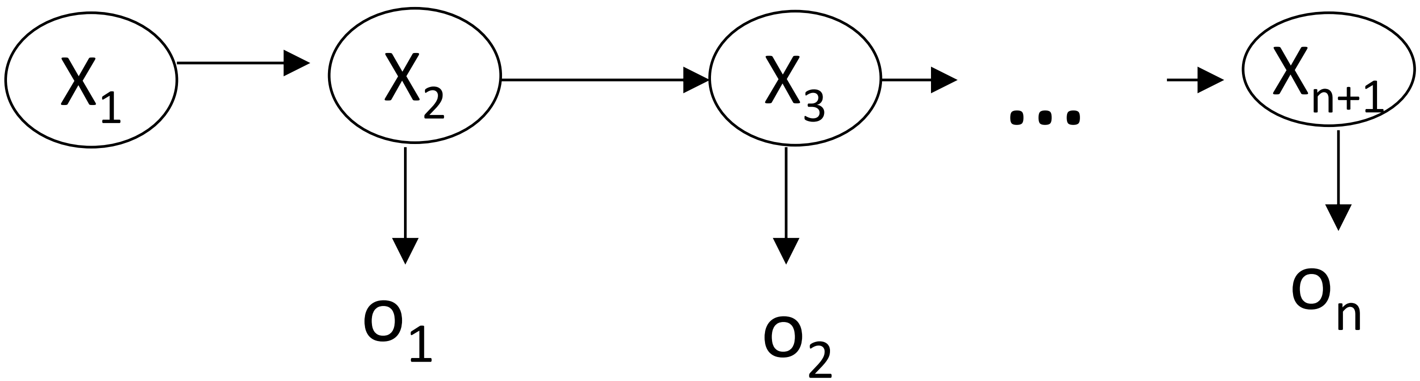


$$P(X,Y) = P(f1, f2, ..., f_n, Y)$$

$$= P(Y)P(f_1|Y)...P(f_n|Y)$$

$$= P(Y)\prod_{k=1}^{n} P(f_k|Y)$$

HMM



- State sequence: $X_{1,n+1}$
- Output sequence: $O_{1,n}$

$$P(O_{1,n}, X_{1,n+1}) = \pi(X_1)\prod_{i=1}^{n}(P(X_{i+1} \mid X_i)P(O_i \mid X_{i+1}))$$

Generative model

- It is a directed graphical model in which the output (i.e., what to be predicted) topologically precede the input (i.e., what is given as observation).

- Naïve Bayes and HMM are generative models.

Markov random field

Markov random field

- Also called “Markov network”

- It is a graphical mode in which a set of random variables have a Markov property:
  - Local Markov property: A variable is conditionally independent of all other variables given its neighbors.

$$P(X_i|X_j, ne(X_i)) = P(X_i|ne(X_i))$$

Cliques

- *A* **clique** in an undirected graph is a subset of its vertices such that every two vertices in the subset are connected by an edge.
- *A* **maximal clique** is a clique that cannot be extended by adding one more vertex.
- *A* **maximum clique** is a clique of the largest possible size in a given graph.

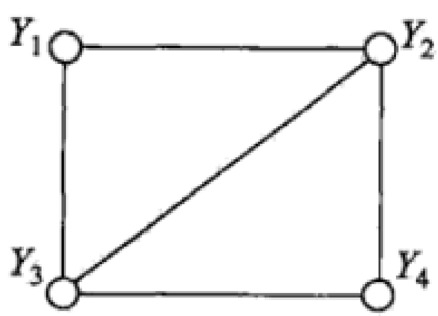


图 11.3 无向图的团和最大团

clique:

maximal clique:

maximum clique:

Clique factorization

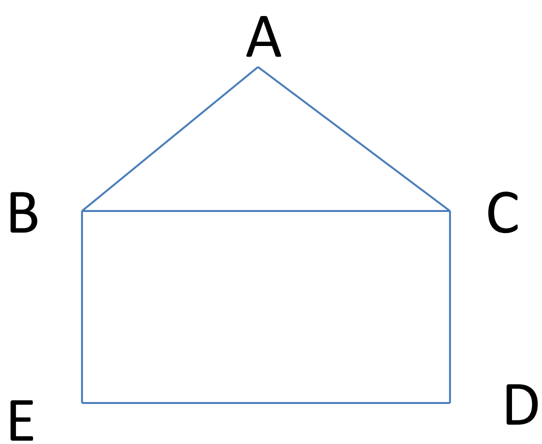
$G = (V, E)$ be an undirected graph.

$cl(G)$ be the set of cliques of G.

$$P(X) = \frac{1}{Z} \prod_{C \in cl(G)} \phi_C(X_C)$$

$\phi_C(X_C)$ 是C上定义的严格正函数

乘积是在无向图所有的最大团上进行的



$$P(A,B,C,D,E)$$

$$= \frac{1}{Z}\phi_{ABC}(A,B,C)\phi_{BE}(B,E)\phi_{DE}(D,E)\phi_{C,D}(C,D)$$

Conditional Random Field

设X与Y是随机变量,$P(Y|X)$ 在给定X的条件下Y的条件概率分布。

若随机变量Y构成一个由无向图$G=(V,E)$表示的马尔可夫随机场，即

$$P(Y_v \mid X, Y_{w,}\ w \neq v) = P(Y_v \mid X, Y_{w,}\ w \sim v)$$

对任意结点v成立，则称条件概率分布$P(Y|X)$为条件随机场。

Y_v 和Y_W为结点v， u与W对应的随机变量,$W\neq v$表结点v以外的所有结点， $W\sim v$表示在图G中与结点v有边连接的所有结点W 。

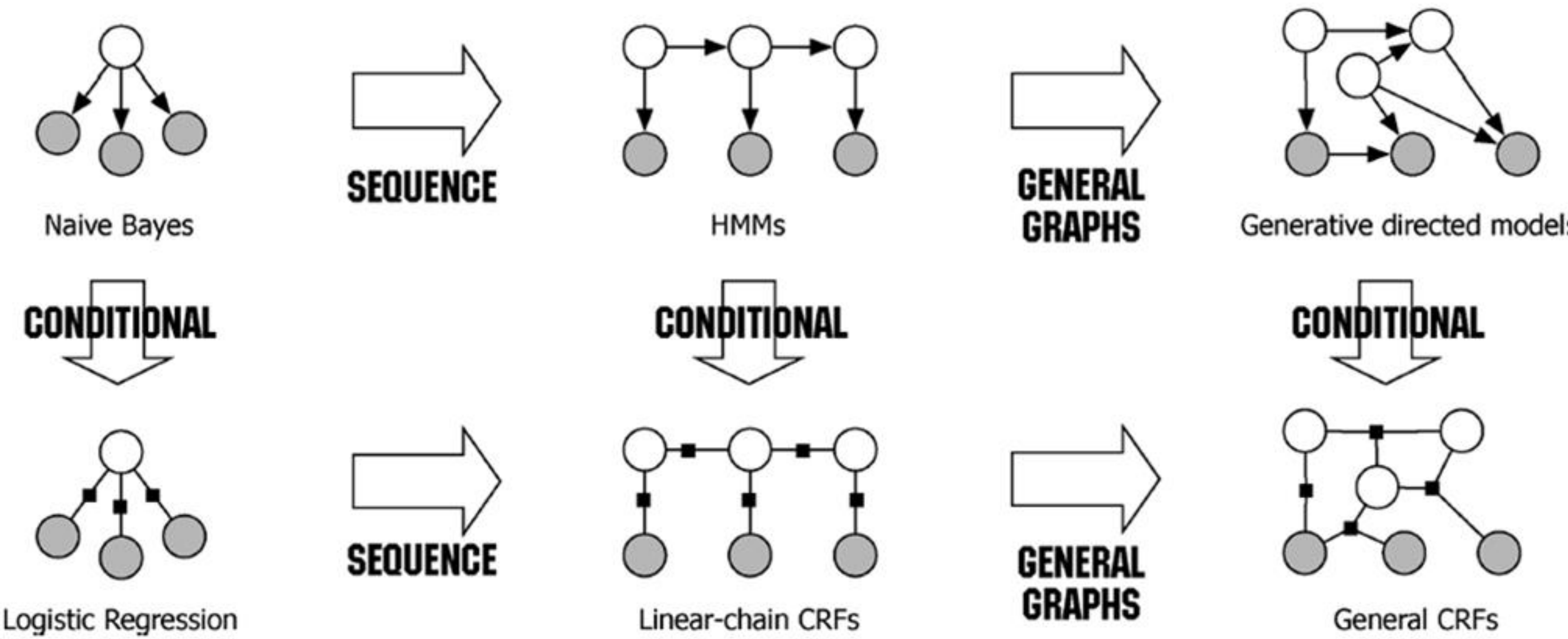
Linear-chain CRF

Motivation

- Sequence labeling problem: e.g., POS tagging
  - HMM: Find best sequence, but cannot use rich features
  - MaxEnt: Use rich features, but may not find the best sequence

- Linear-chain CRF: HMM + MaxEnt

Relations between NB, MaxEnt, HMM, and CRF



Linear-chain CRF

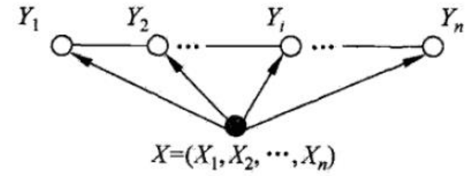


图 11.4 线性链条件随机场

$$f_j(y_{t-1}, y_t, x, t) = \begin{cases} 1 & (y_{t-1} = IN) \wedge (y_t = NNP) \wedge (x_t = Sept) \\ 0 & otherwise \end{cases}$$

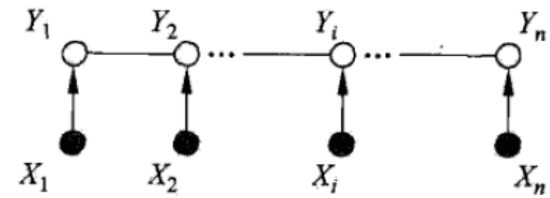


图 11.5 X 和 Y 有相同的图结构的线性链条件随机场

$$F_j(y, x) = \sum_{t=1}^{T} f_j(y_{t-1}, y_t, x, t)$$

$$\begin{aligned} P(y|x) &= \frac{1}{Z(x)} exp(\sum_j \lambda_j F_j(y, x)) \\ &= \frac{1}{Z(x)} exp(\sum_j (\lambda_j \sum_{t=1}^{T} f_j(y_t, y_{t-1}, x, t))) \\ &= \frac{1}{Z(x)} exp(\sum_j \sum_{t=1}^{T} (\lambda_j f_j(y_t, y_{t-1}, x, t))) \\ &= \frac{1}{Z(x)} exp(\sum_{t=1}^{T} \sum_j (\lambda_j f_j(y_t, y_{t-1}, x, t))) \\ &= \frac{1}{Z(x)} \prod_{t=1}^{T} \boxed{exp(\sum_j (\lambda_j f_j(y_t, y_{t-1}, x, t)))} \\ &= \frac{1}{Z(x)} \prod_{t=1}^{T} \phi_t(y_t, y_{t-1}, x) \end{aligned}$$

Training and decoding

$$P(y|x) = \frac{1}{Z(x)} \prod_{t=1}^{T} \phi_t(y_t, y_{t-1}, x)$$

$$\phi_t(y_t, y_{t-1}, x) = exp(\sum_j (\lambda_j f_j(y_t, y_{t-1}, x, t)))$$

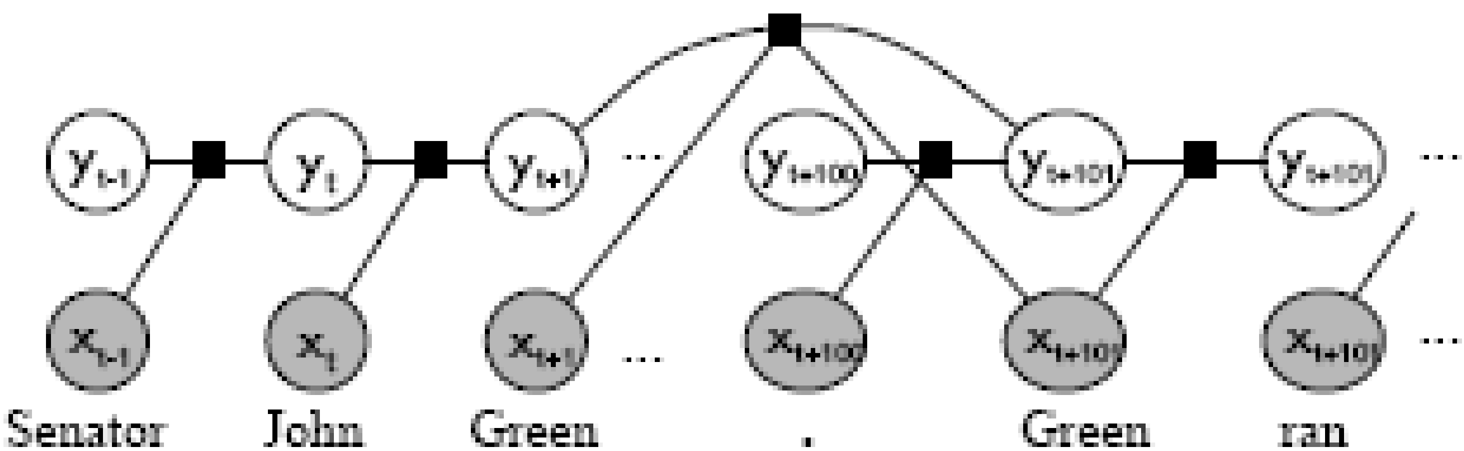
- Training: $\text{estimate } \lambda_j$
  - similar to the one used for MaxEnt
  - Ex: L-BFGS
- Decoding: find the best sequence y
  - similar to the one used for HMM
  - Viterbi algorithm

Skip-chain CRF

Motivation

- Sometimes, we need to handle long-distance dependency, which is not allowed by linear-chain CRF

- An example: NE detection
  - "Senator John Green … Green ran …"



Linear-chain CRF:

$$P(y|x) = \frac{1}{Z(x)} \prod_{t=1}^{T} \phi_t(y_t, y_{t-1}, x)$$

$$\phi_t(y_t, y_{t-1}, x) = exp(\sum_k (\lambda_k f_k(y_t, y_{t-1}, x, t)))$$

Skip-chain CRF:

$$P(y|x) = \frac{1}{Z(x)} \boxed{\prod_{t=1}^{T} \phi_t(y_t, y_{t-1}, x)} \boxed{\prod_{(u,v)\in D} \phi_{uv}(y_u, y_v, x)}$$

$$\phi_t(y_t, y_{t-1}, x) = exp(\sum_k (\lambda_k f_k(y_t, y_{t-1}, x, t)))$$

$$\phi_{uv}(y_u, y_v, x) = exp(\sum_k (\lambda_{2k} f_{2k}(y_u, y_v, x, u, v)))$$

Summary

- Graphical models:
  - Bayesian network (BN)
  - Markov random field (MRF)

- CRF is a variant of MRF:
  - Linear-chain CRF: HMM + MaxEnt
  - Skip-chain CRF: can handle long-distance dependency
  - General CRF

- Pros and cons of CRF:
  - Pros: higher accuracy than HMM and MaxEnt
  - Cons: training and inference can be very slow

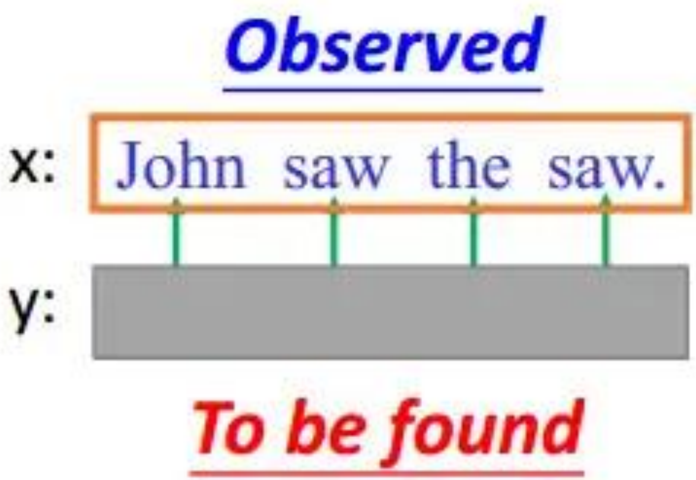
表 12.1 10 种统计学习方法特点的概括总结

方法	适用问题	模型特点	模型类型	学习策略	学习的损失函数	学习算法
感知机	二类分类	分离超平面	判别模型	极小化误分点到超平面距离	误分点到超平面距离	随机梯度下降
k 近邻法	多类分类，回归	特征空间，样本点	判别模型			
朴素贝叶斯法	多类分类	特征与类别的联合概率分布，条件独立假设	生成模型	极大似然估计，极大后验概率估计	对数似然损失	概率计算公式，EM 算法
决策树	多类分类，回归	分类树，回归树	判别模型	正则化的极大似然估计	对数似然损失	特征选择，生成，剪枝
逻辑斯谛回归与最大熵模型	多类分类	特征条件下类别的条件概率分布，对数线形模型	判别模型	极大似然估计，正则化的极大似然估计	逻辑斯谛损失	改进的迭代尺度算法，梯度下降，拟牛顿法
支持向量机	二类分类	分离超平面，核技巧	判别模型	极小化正则化合页损失，软间隔最大化	合页损失	序列最小最优化算法（SMO）
提升方法	二类分类	弱分类器的线性组合	判别模型	极小化加法模型的指数损失	指数损失	前向分步加法算法
EM 算法[①]	概率模型参数估计	含隐变量概率模型		极大似然估计，极大后验概率估计	对数似然损失	迭代算法
隐马尔可夫模型	标注	观测序列与状态序列的联合概率分布模型	生成模型	极大似然估计，极大后验概率估计	对数似然损失	概率计算公式，EM 算法
条件随机场	标注	状态序列条件下观测序列的条件概率分布，对数线性模型	判别模型	极大似然估计，正则化极大似然估计	对数似然损失	改进的迭代尺度算法，梯度下降，拟牛顿法

THANK YOU!

HMM – How to do POS Tagging?

- We can compute P(x,y)



$$y = arg \max_{y \in Y} P(y|x)$$

$$= arg \max_{y \in Y} \frac{P(x,y)}{P(x)}$$

$$= arg \max_{y \in \mathbb{Y}} P(x,y)$$

Task: given x, find y

HMM - Summary

Problem 1: Evaluation

F(x,y)=P(x,y)= P(y)P(x|y)

Problem 2: Inference

$$\tilde{y} = arg \max_{y \in \mathbb{Y}} P(x, y)$$

Problem 3: Training

P(y) and P(x|y) can be simply obtained from training data

HMM的缺点(Drawback)

- 1、只依赖于每一个状态和它对应的观察对象

计算转移概率和输出概率是分开计算的，假设它们是相互独立的。

序列标注问题不仅和单个词相关，而且和观察序列的长度，单词的上下文，等等相关。

- 2、目标函数和预测目标函数不匹配

学到的是状态和观察序列的联合分布P(Y,X)，而预测问题中，需要的是条件概率P(Y|X)。

Feature Vector

- What does $\phi(x, y)$ look like?

x: The dog ate the homework.

y: D N V D N

- $\phi(x, y)$ has two parts
  - Part 1: relations between tags and words
  - Part 2: relations between tags

If there are |S| possible tags, |L| possible words

Part 1 has |S| X |L| dimensions

Part 1	Value
D, the	2
D, dog	0
D, ate	0
D, homework	0
……	……
N, the	0
N, dog	1
N, ate	0
N, homework	1
……	……
V, the	0
V, dog	0
V, ate	1
V, homework	0
……	……

Feature Vector

- What does $\phi(x, y)$ look like?

x: The dog ate the homework.

y: D N V D N

- $\phi(x, y)$ has two parts
  - Part 1: relations between tags and words
  - $\boxed{\text{Part 2: relations between tags}}$

$N_{s,s'}(x, y)$: Number of tags s and s' consecutively in (x, y)

	Part 2	Value
$N_{D,D}(x, y) \rightarrow$	D, D	0
$N_{D,N}(x, y) \rightarrow$	D, N	2
	D, V	0
	……	……
	N, D	0
	N, N	0
	N, V	1
	……	……
	V, D	1
	V, N	0
	V, V	0
	……	……
	Start, D	1
	Start, N	0
	……	……
	End, D	0
	End, N	1

CRF – Training Criterion

$$P(y|x) = \frac{P(x,y)}{\sum_{y'} P(x,y')}$$

- Given training data: $\{(x^1, \hat{y}^1), (x^2, \hat{y}^2), \cdots (x^N, \hat{y}^N)\}$
- Find the weight vector w^* maximizing objective function $O(w)$:

$$w^* = \arg\max_w \mathrm{O}(w) \quad \mathrm{O}(w) = \sum_{n=1}^{N} logP(\hat{y}^n|x^n)$$

$$logP(\hat{y}^n|x^n) = log\underbrace{P(x^n,\hat{y}^n)}_{\text{Maximize what we observe}} - \underbrace{log\sum_{y'} P(x^n,y')}_{\text{Minimize what we don't observe}}$$

Maximize what we observe

Minimize what we don't observe

CRF – Inference

- Inference

$$y = arg \max_{y \in Y} P(y|x) = arg \max_{y \in Y} P(x, y)$$

$$= arg \max_{y \in Y} w \cdot \phi(x, y)$$ Done by Viterbi as well

$$\mathrm{P}(x, y) \propto exp\big(w \cdot \phi(x, y)\big)$$

CRF - Summary

Problem 1: Evaluation

$$F(x,y) = P(y|x) = \frac{exp\big(w \cdot \phi(x,y)\big)}{\sum_{y' \in \mathbb{Y}} exp\big(w \cdot \phi(x,y')\big)}$$

Problem 2: Inference

$$\tilde{y} = \arg\max_{y \in \mathbb{Y}} P(y|x) = \arg\max_{y \in \mathbb{Y}} w \cdot \phi(x,y)$$

Problem 3: Training

$$w^* = \arg\max_{w} \prod_{n=1}^{N} P(\hat{y}^n | x^n)$$

$$\mathrm{w} \rightarrow w + \eta \left(\phi(x^n, \hat{y}^n) - \sum_{y'} P(y'|x^n)\, \phi(x^n, y') \right)$$

- CRF没有HMM那样严格的独立性假设条件，因而可以容纳任意的上下文信息。
- CRF模型解决了标注偏置问题，去除了HMM中不合理的假设。